

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_12apr25_isolado.parquet | Experiment: rssi_odom_dataset_11apr25_isolado.parquet

Params:

- model_type: GRU
- group_sec: 100
- timesteps_orig: 60
- batch_size: 1024
- min_units: 64
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']
- min_layers: 1
- max_layers: 2
- max_epochs: 20
- hyperband_iterations: 3
- group_msec: 100
- samples_per_sec: 10
- forecast_horizon_sec: 3.0
- forecast_steps: 30
- timesteps: 60

Tuning:

Trials executed: 10/10

Training Performance:

Loss: 0.0075 | MAE: 0.0659

Validation Performance:

Loss: 0.0097 | MAE: 0.0764

Direct (multi-output) Performance:

MAE: 1.3272 | MSE: 3.0887 | RMSE: 1.7575 | R2: 0.8814

Multi-Step Performance (aggregate):

Steps: 30 (~3.0 s) | MAE: 0.0399 | MSE: 0.0027 | RMSE: 0.0519 | R2: 0.9434

Multi-Step (denormalized, all steps combined):

MAE: 1.52 | MSE: 4.06 | RMSE: 2.02 | R2: 0.881

Per-step (denormalized):

Step 1 → MAE: 1.06 | MSE: 1.92 | RMSE: 1.38 | R2: 0.944

Step 30 → MAE: 1.74 | MSE: 5.14 | RMSE: 2.27 | R2: 0.850

Best Hyperparameters:

Layers: 2 | Activation: elu | Optimizer: Nadam | Dropout: 0.2

Timing:

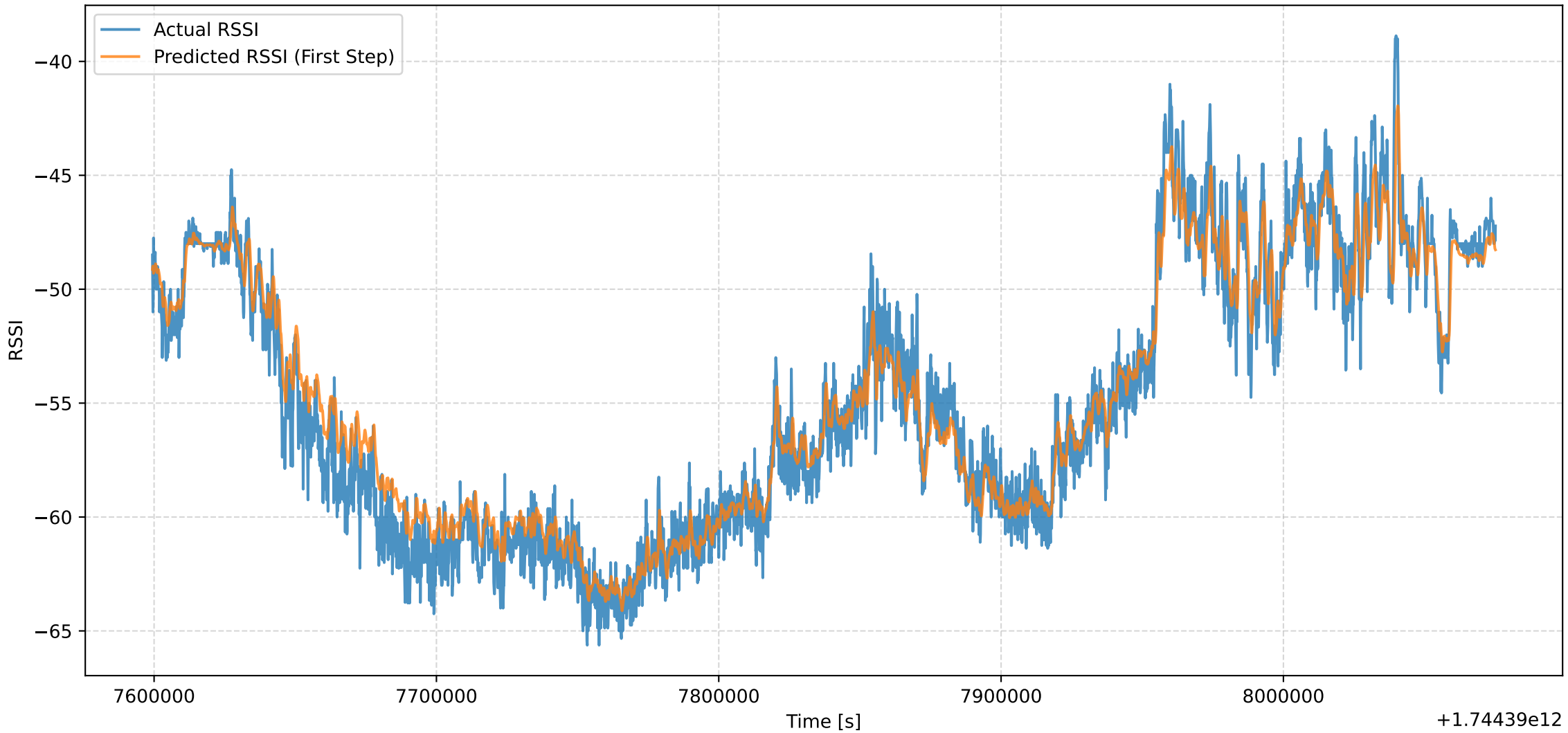
Train: 2025-09-28_08-05-42 → 2025-09-28_08-09-27 (3.76 min)

Predict: 2025-09-28_08-09-27 → 2025-09-28_08-12-31 (3.07 min)

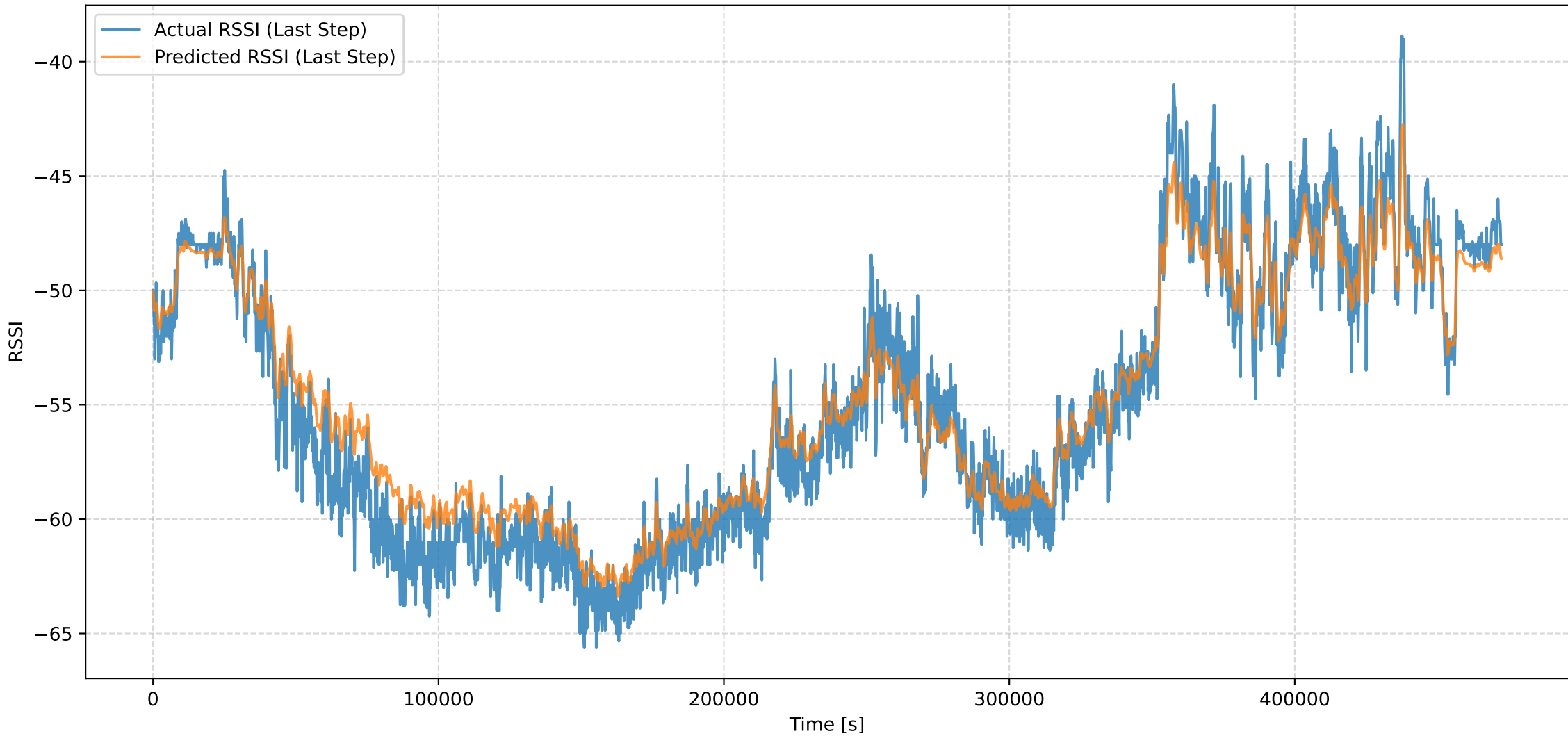
Training dataset down-sample:
Rows before: 31097 | after: 3738

Experiment dataset down-sample:
Rows before: 40318 | after: 4847

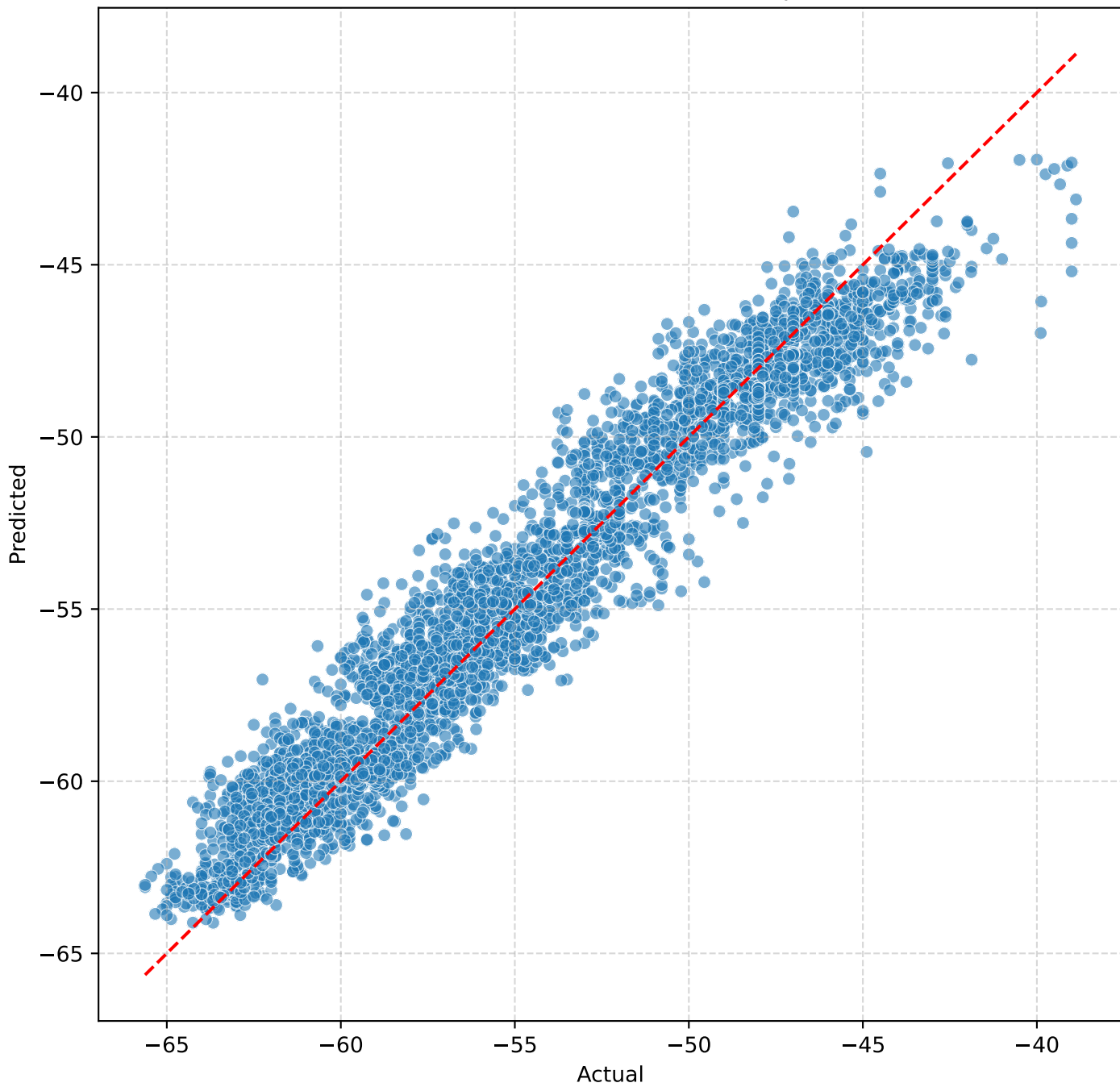
Actual vs Predicted RSSI (First Step)



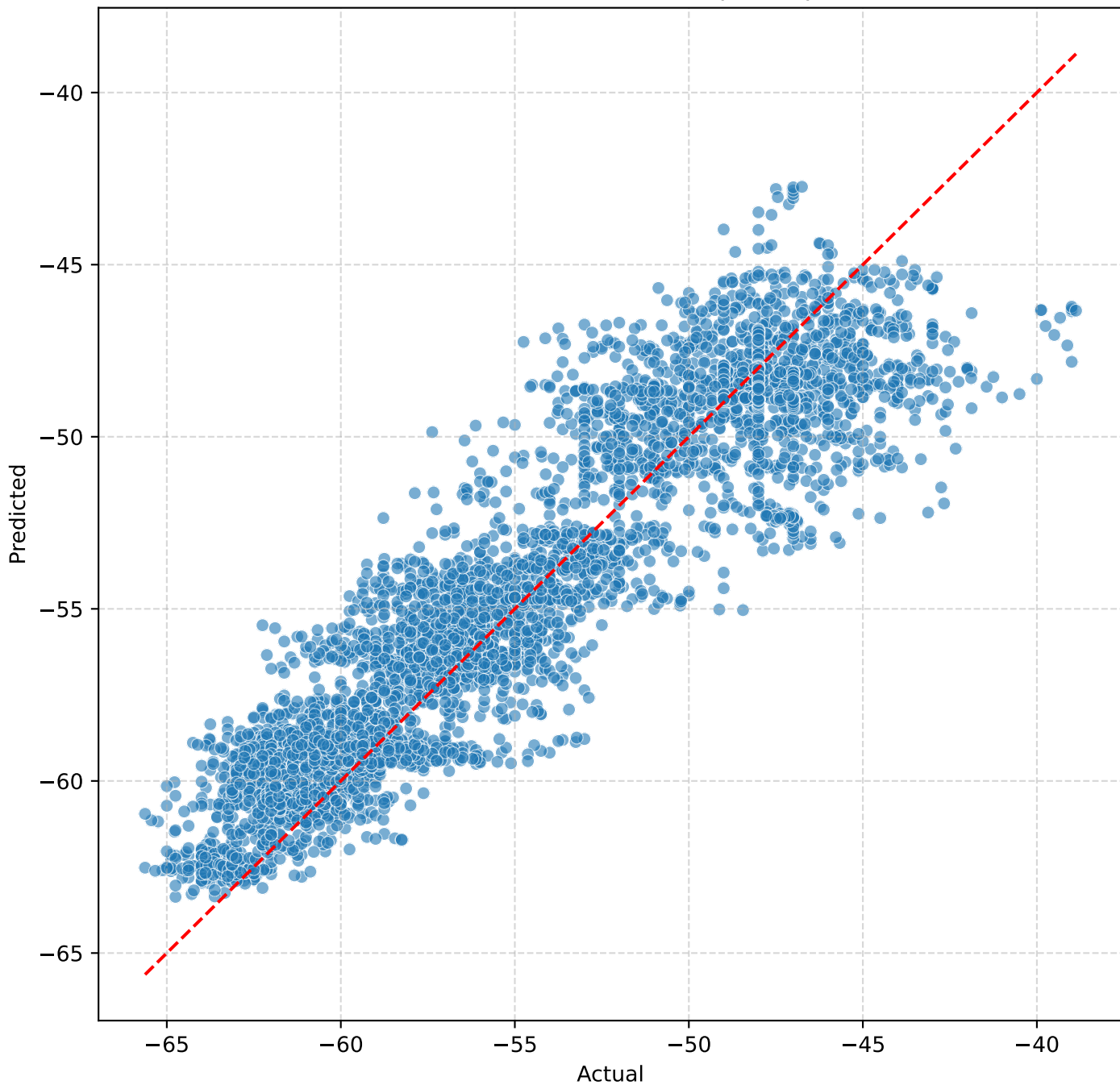
Actual vs Predicted RSSI (Last Step - step 30 $\approx$ 3.0s) [lag=33 samples $\approx$ 3.30s]



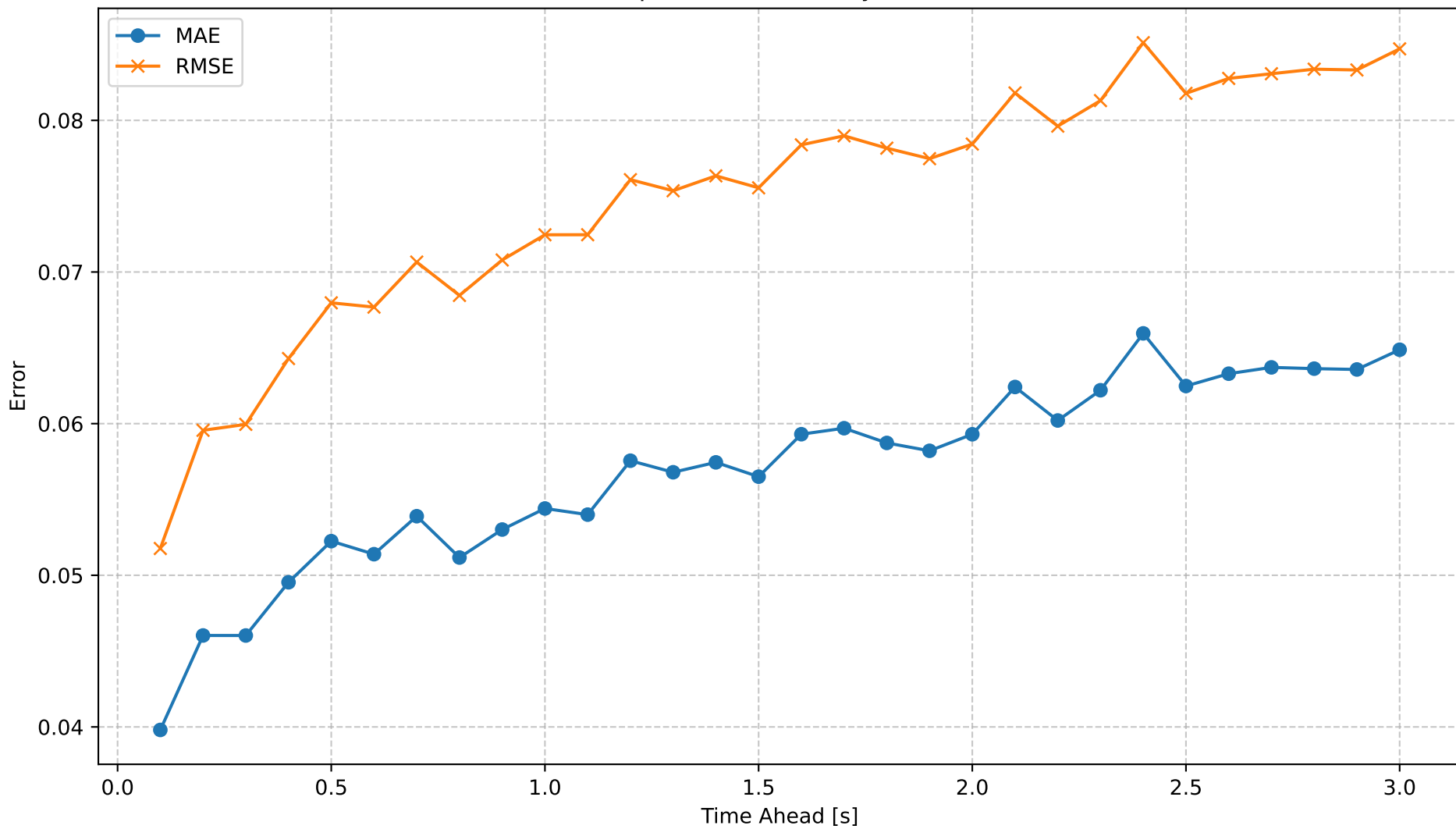
Actual vs Predicted (First Step)



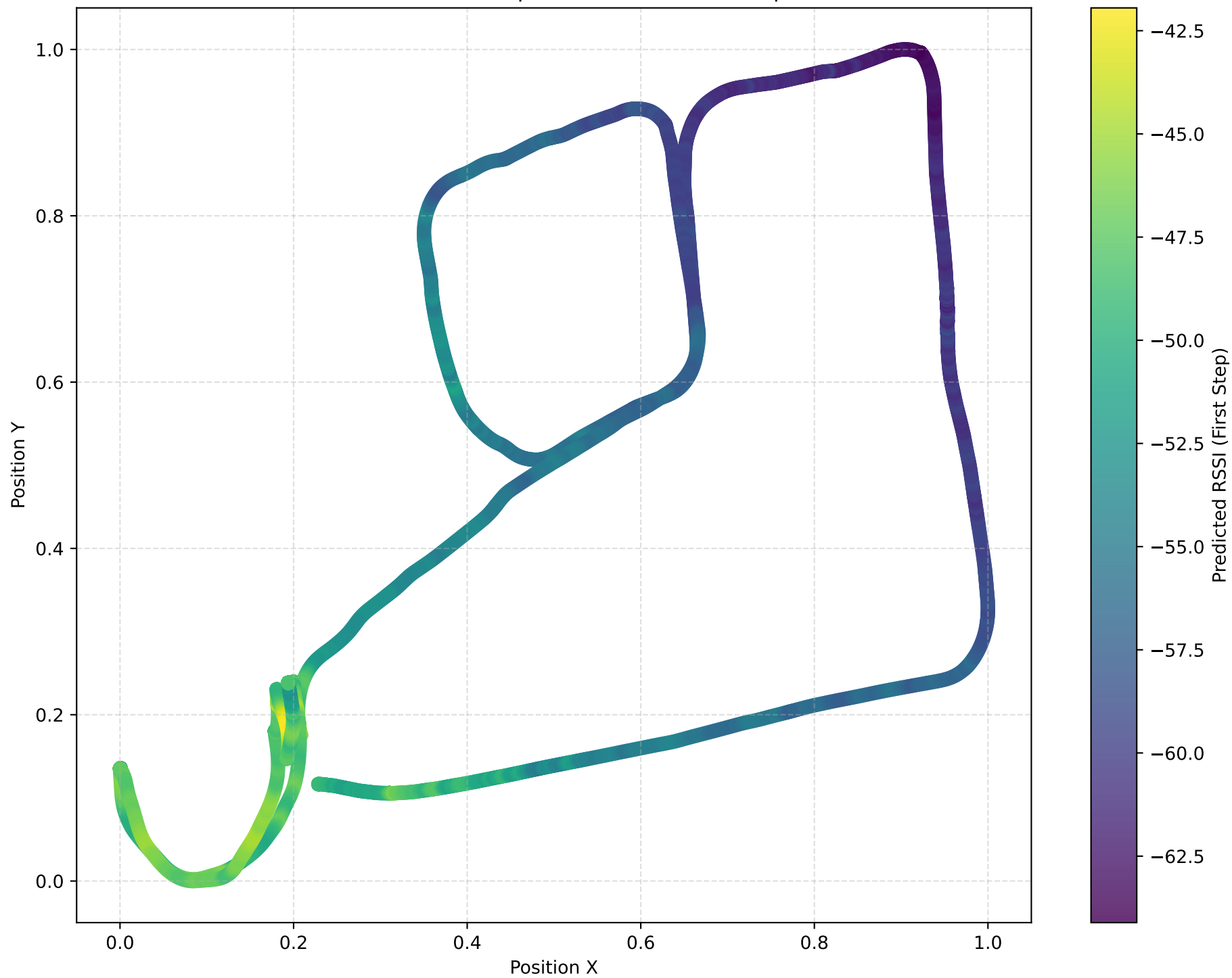
Actual vs Predicted (Last Step - step 30)



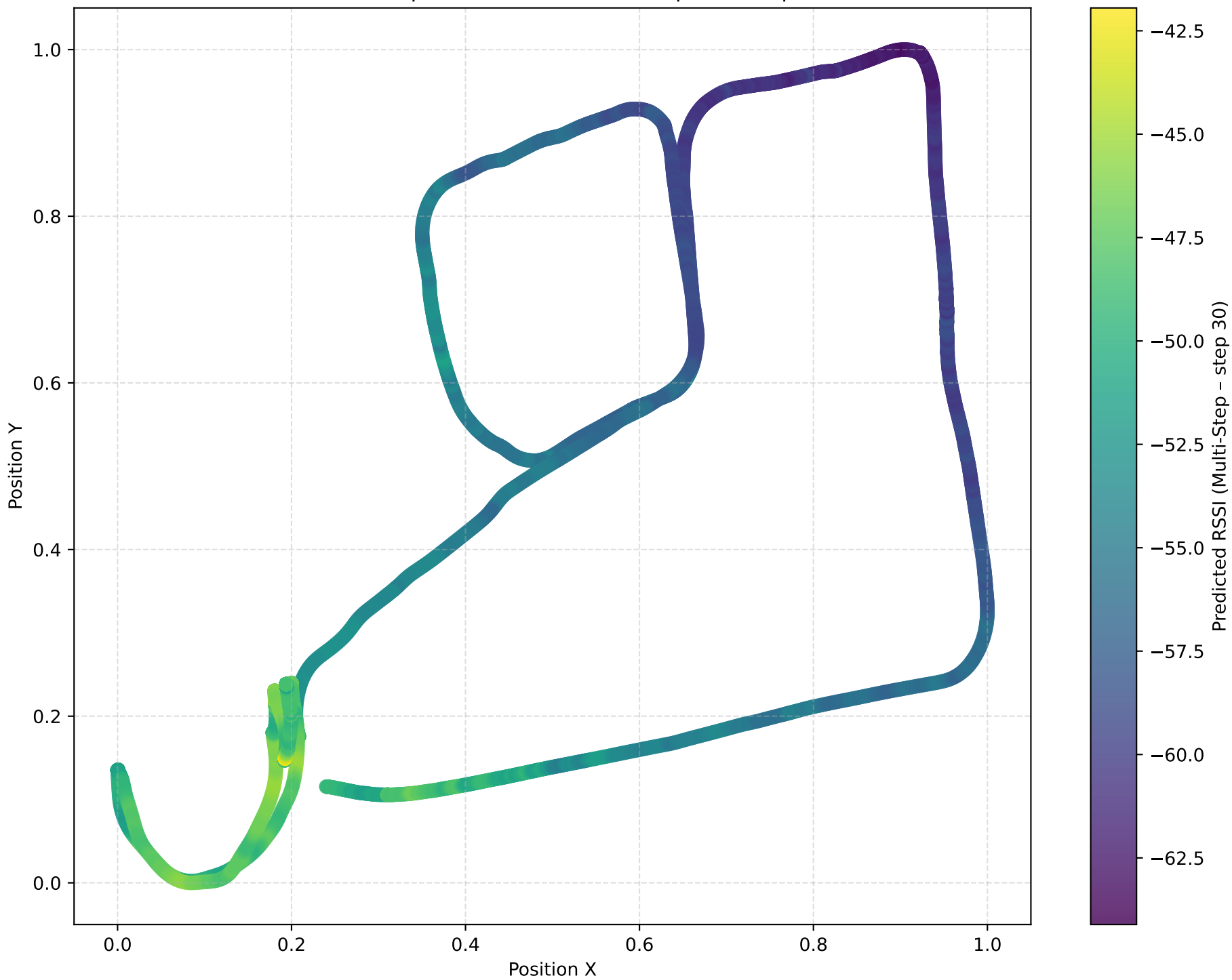
Multi-Step Forecast Error by Time Ahead



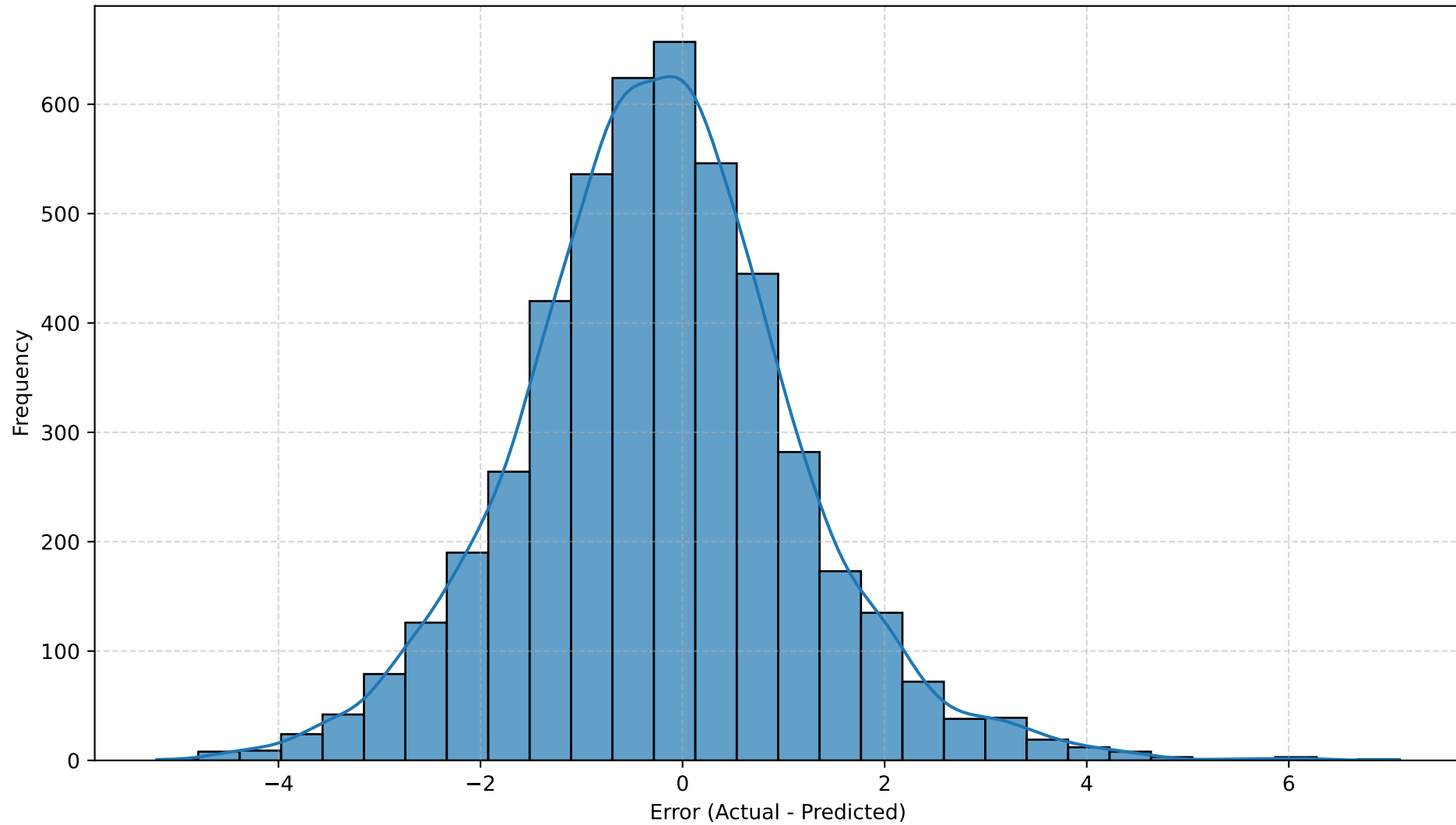
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - 30 steps $\approx$ 3.0 s)



Distribution of Prediction Errors (First Step)



Distribution of Prediction Errors (Last Step - step 30)

